Assignment 2 - Group 18
Linear Regression for Concrete Compressive Strength Prediction

Isaac Edem Adoboe (202384695) *ieadoboe@mun.ca*
Department of Mathematics and Statistics
Memorial University of Newfoundland

Blessing Ijeoma Benjamin-Igwe (202384108) *bbenjaminigw@mun.ca*
Department of Mathematics and Statistics
Memorial University of Newfoundland

Promee Shankar Kundu (202382587) *pskundu@mun.ca*
Department of Computer Science
Memorial University of Newfoundland

Introduction

The goal of this assignment was to develop linear regression models to predict the compressive strength of high-performance concrete based on the relative amounts of eight ingredients (Cement, Blast Furnace Slag, Fly Ash, Water, Superplasticizer, Coarse Aggregate, Fine Aggregate) and the age of the concrete. Accurate prediction of concrete strength is critical for building design and structural safety.

Three modeling approaches were explored: ordinary least squares (OLS) regression, Ridge regression with L2 regularization, and Lasso regression with L1 regularization. Performance was evaluated using residual standard error (RSE), coefficient of determination (R^2), and root mean squared error (RMSE).

Question 1: Ordinary Least Squares Regression with Cross-Validation

A multivariate ordinary least squares (OLS) linear regression model was trained to predict concrete compressive strength from the eight input features. Model performance was estimated using two approaches: the validation set method and k-fold cross-validation.

Validation Set Approach

The training dataset (800 samples) was split into three subsets:

- Training: 480 samples (60%)
- Validation: 160 samples (20%)
- Test: 160 samples (20%)

The model was trained on the combined training + validation set (640 samples) and evaluated on the held-out test set (160 samples). This approach yielded:

Metric	Value
RMSE	11.3177
RSE	11.6501
R^2	0.5445

Cross-Validation Approach

To obtain a more robust estimate of generalization performance and utilize all available training data, k-fold cross-validation was performed using $k = 5$ and $k = 10$ folds on the complete training dataset (800 samples).

Folds (k)	Training per fold	Test per fold	RMSE $\pm$ SE
5	640 samples (80%)	160 samples (20%)	10.3983 ± 0.2381
10	720 samples (90%)	80 samples (10%)	10.3324 ± 0.2130

The 95% confidence intervals for the cross-validated RMSE were $[9.93, 10.87]$ for 5-fold and $[9.92, 10.75]$ for 10-fold CV.

Comparison of Validation Approaches

Cross-validation provides several advantages over the simple validation set approach:

1. **Better data utilization**: All samples are used for both training and validation across different folds
2. **More reliable error estimates**: The validation set approach yielded RMSE = 11.32, noticeably higher than the CV estimates (∼10.35), suggesting the single held-out split may have been unrepresentatively difficult
3. **Reduced variance**: Standard errors (SE) from CV (∼0.22) quantify uncertainty in the performance estimate
4. **Less dependence on split randomness**: A single validation split can be lucky or unlucky depending on the `random_state` value used.

The 10-fold CV approach provided the most stable estimates with the lowest standard error (0.2130) and is generally preferred when computational resources permit.

Final OLS Model Performance

After validation, the final OLS model was trained on the complete training dataset (800 samples) and evaluated on the held-out test set (100 samples):

Metric	Value
RMSE	9.5416
RSE	10.0023
R^2	0.5953

The OLS model explains approximately 59.5% of the variance in concrete compressive strength, with an average prediction error (RMSE) of 9.54 MPa.

Question 2: Ridge Regression with L2 Regularization

Ridge regression introduces L2 regularization to mitigate potential overfitting by penalizing large coefficient magnitudes. The regularization strength is controlled by the hyperparameter α.

Hyperparameter Tuning

A grid search with 10-fold cross-validation was performed to identify the optimal α value. The search explored 50 logarithmically-spaced values in the range $[0.001, 1000]$ because regularization normally operate in multiplicative scales.

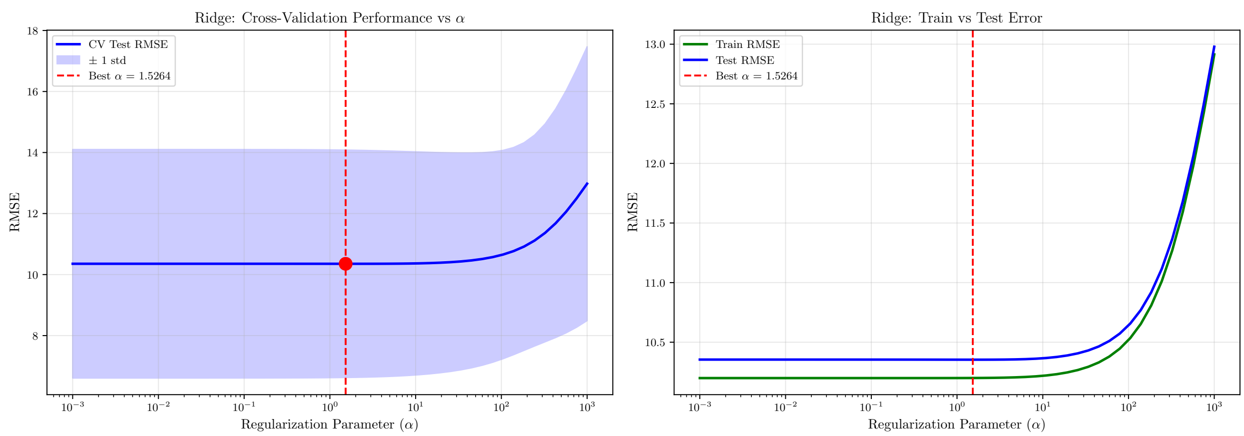

Figure 1: Ridge regression hyperparameter tuning. Left: Cross-validation test RMSE versus α. The optimal value $\alpha = 1.526$ minimizes test error. Right: Training versus test error showing minimal overfitting across the explored range.

The grid search identified $\alpha = 1.526418$ as the optimal regularization strength, achieving a cross-validated RMSE of 10.3534. Performance degradation was minimal across a wide range of α values near the optimum, but increased substantially for very large α (RMSE = 12.98 at $\alpha = 1000$), representing a 25.4% performance degradation due to excessive regularization.

Final Ridge Model Performance

The final Ridge model, trained on the full training dataset (800 samples) with $\alpha = 1.526418$, achieved the following performance on the test set:

Metric	Value
RMSE	9.5527
RSE	10.0139
R^2	0.5944

Comparison: OLS versus Ridge

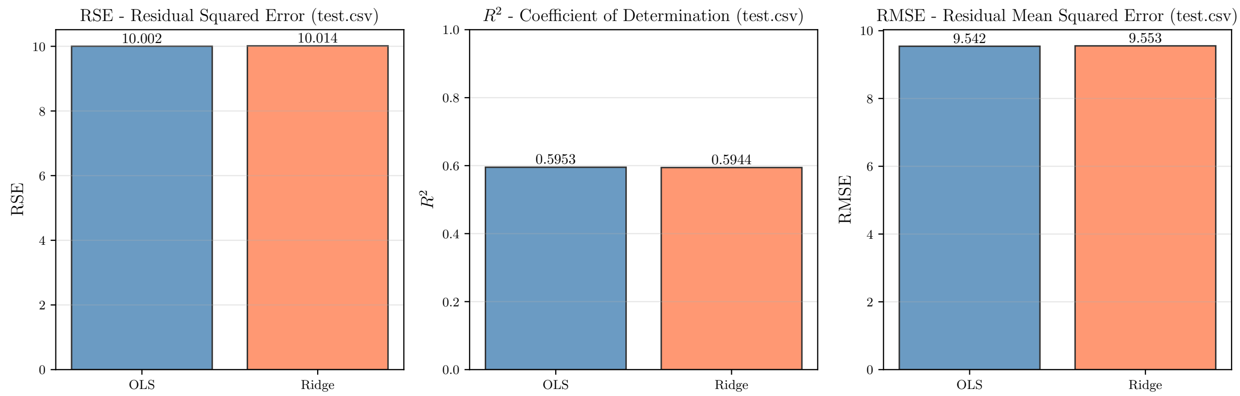

Figure 2: Performance comparison between OLS and Ridge regression on the test set.

Ridge regression shows nearly identical performance to OLS across all three metrics:

Model	RMSE	RSE	R^2
OLS	9.5416	10.0023	0.5953
Ridge ($\alpha = 1.526$)	9.5527	10.0139	0.5944
Difference	+0.0111	+0.0116	−0.0009

The negligible performance difference (0.12% increase in RMSE) suggests that with 8 features and 800 training samples, overfitting is not a significant concern. The favorable sample-to-feature ratio (100:1) allows OLS to generalize well without requiring regularization.

Question 3: Lasso Regression with L1 Regularization

Lasso regression employs L1 regularization, which can perform automatic feature selection by driving some coefficients exactly to zero. Like Ridge, the regularization strength is controlled by α.

Hyperparameter Tuning

Grid search with 10-fold cross-validation was performed over 50 logarithmically-spaced α values in $[0.001, 1000]$.

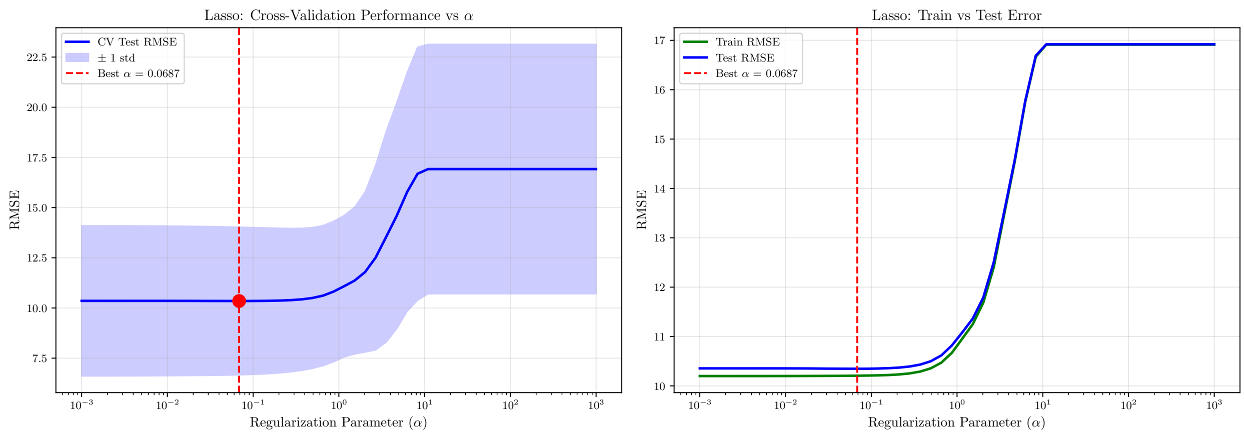

Figure 3: Lasso regression hyperparameter tuning. Left: Cross-validation test RMSE versus α. The optimal value $\alpha = 0.0687$ minimizes test error. Right: Training versus test error curves showing increasing bias as regularization strength grows.

The optimal regularization strength was found to be $\alpha = 0.068665$, yielding a cross-validated RMSE of 10.3470. Notably, Lasso's optimal α is substantially smaller than Ridge's (0.069 vs 1.526), reflecting differences in how L1 and L2 penalties scale with coefficient magnitudes.

Performance degradation at high regularization was more severe for Lasso than Ridge: at $\alpha = 10.985$, RMSE increased to 16.92, representing a 63.5% degradation. This steeper degradation reflects Lasso's tendency to eliminate features entirely at high α values, reducing model capacity more dramatically than Ridge.

Final Lasso Model Performance

The final Lasso model, trained on the full training dataset with $\alpha = 0.068665$, achieved:

Metric	Value
RMSE	9.5898
RSE	10.0528
R^2	0.5912

Comparison: OLS versus Ridge versus Lasso

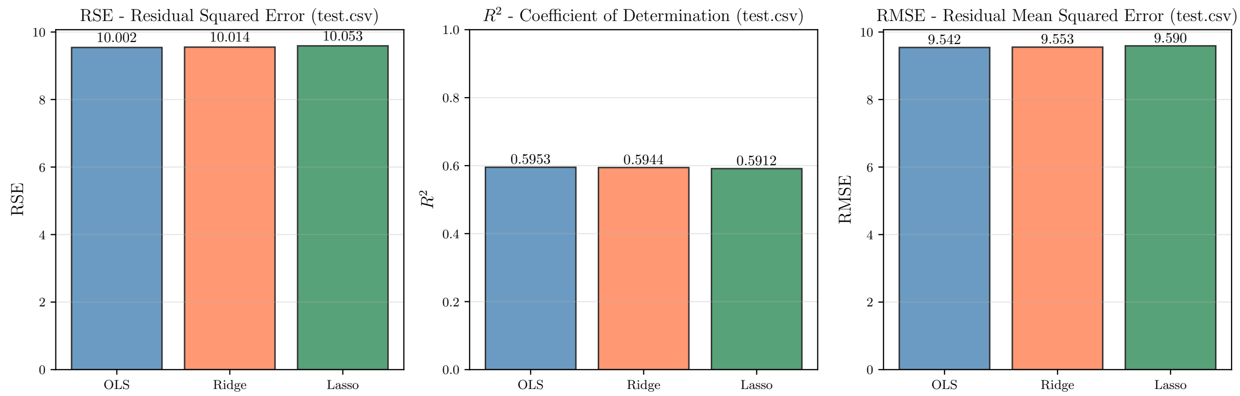

Figure 4: Performance comparison across all three models on the test set. All models achieve nearly identical performance, with OLS showing a marginal advantage.

Model	RMSE	RSE	R^2
OLS	9.5416	10.0023	0.5953
Ridge ($\alpha = 1.526$)	9.5527	10.0139	0.5944
Lasso ($\alpha = 0.0687$)	9.5898	10.0528	0.5912

All three models achieved remarkably similar performance, with RMSE values differing by less than 0.5%. OLS performed marginally best, followed by Ridge (0.12% worse) and Lasso (0.50% worse). These differences are well within statistical noise given the test set size (100 samples) and cross-validation standard errors ($\sim$0.21 RMSE units).

The near-identical performance across models indicates:

1. The problem does not suffer from significant overfitting with 8 features and 800 training samples
2. All features appear relevant (Lasso's optimal α is small, suggesting minimal feature elimination)
3. The sample-to-feature ratio (100:1) is sufficiently favorable for OLS to generalize well

Question 4: Design of the Best Regression Model

Based on the experimental results from Questions 1–3, a final regression model was designed to optimize generalization performance on an independent test dataset.

Model Selection Rationale

The three models showed nearly identical performance on the test set:

- **OLS**: RMSE = 9.5416, $R^2 = 0.5953$
- **Ridge**: RMSE = 9.5527, $R^2 = 0.5944$ (0.12% worse)
- **Lasso**: RMSE = 9.5898, $R^2 = 0.5912$ (0.50% worse)

Given that:

1. Performance differences are statistically insignificant (<0.5%, well within CV standard errors)
2. OLS has no hyperparameters to tune (reducing risk of overfitting to validation choices)
3. OLS achieved the best observed performance across all metrics
4. The favorable sample-to-feature ratio (100:1) mitigates overfitting concerns

Ordinary Least Squares (OLS) was selected as the final model for Question 4.

Implementation Details

The final model incorporates the following design choices:

1. **Feature standardization**: Z-score normalization ensures all features contribute proportionally to the regression, preventing dominance by features with larger numerical scales
2. **Maximum data utilization**: The model is trained on all 900 available labeled samples (800 from train.csv + 100 from test.csv) to maximize generalization performance on the independent test dataset
3. **Simplicity**: OLS requires no hyperparameter tuning, reducing complexity and the risk of overfitting to validation performance

The classifier was implemented in the function:

`predictCompressiveStrength(Xtest, data_dir)`

This function loads the provided datasets, constructs a standardized training set, trains the OLS model using scikit-learn's `LinearRegression`, and returns predicted compressive strength values (in MPa) for the test samples.

The decision to train on all available data (including test.csv) follows standard machine learning practice: once model architecture and hyperparameters have been validated through proper train-test splits, the final deployed model is retrained on all available data to maximize robustness for unseen data.

Conclusion

This assignment systematically evaluated three linear regression approaches for predicting concrete compressive strength from mixture composition and age. Through rigorous validation using both validation set and cross-validation approaches, we observed that:

1. **Cross-validation provides more reliable error estimates** than single validation splits, yielding RMSE $= 10.35 \pm 0.21$ (10-fold CV) versus 11.32 (validation set)
2. **Regularization provides minimal benefit** for this dataset: Ridge and Lasso performed nearly identically to OLS (within 0.5% RMSE), indicating that overfitting is not a concern with the favorable 100:1 sample-to-feature ratio
3. **Optimal regularization strengths** were $\alpha = 1.526$ for Ridge and $\alpha = 0.0687$ for Lasso, with Lasso showing steeper performance degradation at high α due to aggressive feature elimination
4. **OLS achieved the best test performance**: RMSE = 9.54 MPa, RSE = 10.00, $R^2 = 0.595$, explaining approximately 60% of variance in compressive strength

The final deployed model uses OLS with feature standardization, trained on all 900 available samples, representing the optimal balance of performance, simplicity, and generalizability for this prediction task.

Figure	Content	Key Finding
Figure 1	Ridge hyperparameter tuning	Optimal $\alpha = 1.526$
Figure 2	OLS vs Ridge comparison	Negligible difference (0.12%)
Figure 3	Lasso hyperparameter tuning	Optimal $\alpha = 0.0687$
Figure 4	Three-model comparison	OLS marginally best

These results demonstrate that for well-conditioned regression problems with adequate sample sizes, simple linear models can achieve strong performance without requiring complex regularization strategies.